

Swarn Shekhar

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Summary

Software engineering student with a strong DSA foundation and builder mindset across AI systems, MCP servers, RAG pipelines, computer vision, and full-stack products. Comfortable taking ideas from prototype to deployed, usable systems.

Education

- **Galgotia College Of Engineering And Technology** Gr. Noida
- *B.Tech - Information Technology* **CGPA: 6.8** 2023 - 2027
- **Satyam International School** Patna
- *Class XII - 81.2%* 2022

Experience

- Amazon ML Challenge** Jan 2025 - Feb 2025 **Team Leader**
- Machine Learning, NLP, Computer Vision, XGBoost, Neural Networks*
- Built a **multimodal price prediction system** by combining product text and image signals across **75,000+ Amazon listings**;
 - Pipeline: raw product data → **BERT embeddings** + **CNN image features** → **XGBoost** fusion model → price prediction.

Projects

- ReszVault** [Link](#) ◇
- TypeScript, Bun, Express, React, Prisma, PostgreSQL, pgvector, LangChain, Groq, Better Auth*
- Built a full-stack **PDF intelligence workspace** with auth, PDF upload, indexing, source-grounded chat, citations, and real-time **SSE token streaming**.
 - Engineered RAG: PDF parsing → recursive chunking → **gte-small embeddings** → pgvector search → LLM answer generation.
 - Added authenticated document access, file-size validation, off-topic filtering, and book-level retrieval isolation.
- OVisionMCP** [Link](#) ◇
- Python, MCP, FastMCP, OpenRouter API, Vision Models, Base64, Cross-platform Screen Capture*
- Built an **MCP vision server** that lets text-only LLM clients analyze local images, remote image URLs, and live screenshots through external vision models.
 - Exposed **NoEyeVision** for image reasoning and **screen** for screenshot capture with image path, width, and height metadata.
 - Pipeline: image path/URL → base64 or remote payload → OpenRouter vision request → structured MCP response.
- AlgoLens** [Link](#) ◇
- Python, FastAPI, Streamlit, scikit-learn, Pandas, Graphviz, Decision Trees*
- Built a **no-code ML playground** where users upload CSV datasets, choose targets, tune tree hyperparameters, and train models interactively.
 - FastAPI backend handles CSV ingestion, categorical encoding, target validation, **DecisionTreeClassifier** training, and Graphviz SVG generation.
 - Streamlit frontend visualizes dataset previews, class distribution, parameter controls, learning notes, and rendered decision-tree diagrams.
- WaveNET** [Link](#) ◇
- Go, UDP Broadcast, TCP Relay, Flood Algorithm, TTL, Concurrency, Mutex, net.Pipe, LAN Networking*
- Implemented **LAN peer discovery** using UDP broadcast with live peer tracking, seed-node fallback, and background peer cleanup loops.
 - Built **TCP-based message relay** with flood propagation, TTL hop limits, duplicate-message suppression, path tracing, and concurrency-safe seen-state.
 - Added **gateway delivery simulation**, **live propagation dashboard hooks**, **structured logs**, **unit tests**, and connection-path tests using `net.Pipe()`.

Technical Skills

- **Languages:** C, C++, Python
- **AI / ML:** PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, YOLO, CNN-LSTM, Decision Trees, Feature Engineering
- **LLM / RAG:** LangChain, MCP, OpenRouter API, Groq, Embeddings, Vector Search, pgvector, Prompt Guardrails, SSE Streaming
- **Backend:** FastAPI, Node.js, Express.js, Bun, REST APIs, Auth, File Uploads, API Integration
- **Databases / Cloud:** Supabase, MongoDB, Vercel, Render, AWS

Achievements

- Ranked **Top 5% among 83,000+** participants in the **Amazon ML Challenge 2025**.
- Secured **4th rank** in Galgotias Programming League; solved **1500+ DSA problems** across platforms, including **800+** on LeetCode.

Volunteer Experience

- **Community Lead, Developer Student Clubs (GDC) - Technical Lead, HackBase Club**
- *Workshops for hundreds of students; organized hackathons and coding events for 500+ participants.* Jan 2024 - Present